

Solutions and R Programming Examples: Confidence Interval

1.Fast Food

1.1) 7.093 1.2) 0.363 1.3) 0.779

1.4) (6.315,7.872)

We are 95% confident that the population mean the amount spent for lunch at a fast-food restaurant is between \$6.315 and \$7.872.

R Program

```
food<-c(7.42,6.29, 5.83, 6.50,8.34, 9.51,7.10, 6.80,5.90, 4.89,6.50, 5.52, 7.90, 8.30, 9.60)
```

```
m<- mean(food)
```

```
m
```

```
[1] 7.093333
```

```
sd<-sd(food)
```

```
n<-15
```

```
se<- sd/sqrt(n)
```

```
se
```

```
[1] 0.3630357
```

```
moferr<-qt(0.975,n-1)*se
```

```
moferr
```

```
[1] 0.7786341
```

```
lcl<-m-moferr
```

```
ucl<-m+moferr
```

```
lcl
```

```
[1] 6.314699
```

For $n=15$ and $\alpha=0.05$ the critical value is 2.131

ucl

[1] 7.871967

.....

2. The complaints

2.1) 43.040 2.2) (31.125, 54.955)

.....

3. Flu shot

3.1) 0.00615 3.2) 0.00125 3.3) 0.0025 3.4) (0.0037, 0.00861) 3.5) Yes

.....

4. Basic life

4.1) 0.54255, 0.61381 4.2) (-0.12712, -0.01540)

R Program

```
prop.test(x=c(306,391), n=c(564,637),conf.level=0.95)
```

.....

2-sample test for equality of proportions with continuity correction

data: c(306, 391) out of c(564, 637)

X-squared = 5.9483, df = 1, p-value = 0.01473

alternative hypothesis: two.sided

95 percent confidence interval:

-0.12878962 -0.01373351

sample estimates:

prop 1 prop 2

0.5425532 0.6138148

5. Study time

5.1) 2.567 5.2) (-3.110, 8.244) We are 95% confident that the population mean study time per week between the two groups (students who planned to go to graduate school and those who did not) is between -3.110 – 8.244 hours.

This means that there is no statistically significant difference in the mean study time per week between the two groups.

R Program

```
plan<-c(15,7,15,10,5,5,2,3,12,16,15,37,8,14,10,18,3,25,15,5,5)
```

```
N_plan<-c(6,8,15,6,5,14,10,10,12,5)
```

```
t.test(plan,N_plan,conf.level=0.95,var.equal=TRUE)
```

Two Sample t-test

data: plan and N_plan

t = 0.92469, df = 29, p-value = 0.3628

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

-3.110283 8.243616

sample estimates:

mean of x mean of y

11.66667 9.10000

6. Multiple myeloma

6.1) 86.143 6.2) (-28.261, 200.547) We are 95% confident that the population mean difference in bone marrow microvessel density of patients who received transplants of stem cells (before and after) is between -28.261, 200.547.

This means that there is no statistically significant difference in the mean of bone marrow microvessel density between the two groups.